

LEAN VERIFICATION RECORD

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Result: $K(\omega + 1, \text{ord})$ is second-countable and Lindelöf, and is not σ -compact.

Checked declaration: VietorisOmegaOne.order_model_theorem.

The theorem has no mathematical hypotheses. Its type is `CompactPowerClaim` applied to the extended natural numbers with their order topology.

The carrier consists of sequences with compact image. The topology is induced from the topology generated by a single open range bound and finitely many open coordinate bounds. It matches Definitions 1.8 and 1.14 of Caruvana and Holshouser, arXiv:2507.17936v3, pp. 4-5; the relevant question is Question 1 on p. 11.

Compiler: Lean 4.19.0. Library: mathlib v4.19.0.

mathlib commit: c44e0c8ee63ca166450922a373c7409c5d26b00b.

Verification date: 2026-09-27. Final audit compiler exit code: 0.

Reported transitive axioms: `propext`, `Classical.choice`, `Quot.sound`. The audit contains no `sorryAx`, no custom assumed mathematical result, and no `Lean.ofReduceBool` dependency.

The full local proof sources and dependency lockfile are supplied in the accompanying archive. `FullAudit.lean` checks the exact theorem type and prints its axioms. Run the supplied `verify.ps1` script in PowerShell to rebuild the modules sequentially and repeat the audit.

The machine-readable `verification-status.json` records SHA-256 hashes. These hashes identify the checked source files; the compiler check and the axiom report provide the verification evidence.

Audit log SHA-256:

d615e845f3cc2af273a10441edb099f40cb2dfa5ebbf51d04db6586e91c66bb4

Scope. This is a reproducible local machine-verification record, not an institutional certificate, independent peer-review report, or priority claim. The identification of the arXiv notation with the formal construction is documented in the article and README. The order-topology transfer of the construction is included in the checked code.

AI disclosure. ChatGPT Astra did almost all of the mathematical, formalization and document work. Lucas Alves de Almeida reports that he checked and edited the paper.